BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(END SEMESTER EXAMINATION)

CLASS: BTECH
BRANCH: CS/IT

SEMESTER: VI
SESSION: SP/2024

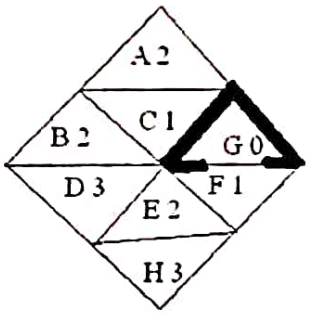
SUBJECT: CS335 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TIME: 3 Hours

FULL MARKS: 50

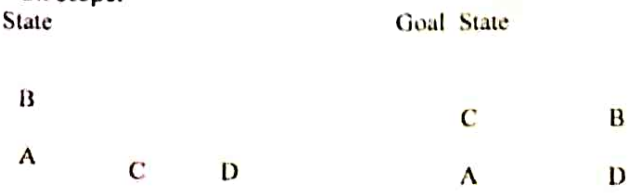
INSTRUCTIONS:
1. The question paper contains 5 questions each of 10 marks and total 50 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.

			CO	BL
Q.1(a)	Describe the differences and similarities between problem solving and planning. Which of the following are true and which are false? Explain your answers. (i) h(n) = 0 is an admissible heuristic for the 8-puzzle. (ii) Depth-first search always expands at least as many nodes as A* search with an admissible heuristic.	[3+2=5]	1,3	4,5
Q.1(b)	Consider the following path-finding problem. One can move from one small triangle to another if they share a vertex (e.g., A can go to B and C). However, the goal G can only be accessed from F. The number after the letter is the heuristic function value for that state. The actual cost of each move is as follows: • A move down one level (e.g. A→C or B→E) costs 1 • A move sideways on the same level (e.g. C→B or E→F) costs 2 • A move up one level (e.g. B→A or C→A) costs 3	[5]	1	3



Perform A* Search, starting from A. Break ties alphabetically. Show your search tree, and circle states that are expanded. What is the cost of your solution path?

Q.2(a)	Describe the basic component of a knowledge base Artificial Intelligence system. Distinguish between Forward and backward chaining.	[2+3=5]	2	4
Q.2(b)	Consider the following sentences: (i) All people that are not poor and are smart are happy. (ii) Those people that read are not stupid. (iii) John can read and is wealthy. (iv) Happy people have exciting lives. Translate the above sentences into formulas in predicate logic. Convert the formulas into clause form. Prove that "Can anyone be found with an exciting life?" using resolution.	[5]	2	3
Q.3(a)	Describe different components of planning system. Illustrate and justify mathematical interpretation of Bayesian Netwok.	[2+3=5]	3	3
Q.3(b)	Devise a plan of action for following "block word problem" using Goal stack method. Explicitly show all steps.	[5]	3	3



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Q.4(a) How the concepts of bias and variance relate to overfitting and underfitting? [2+3=5] 4 3,5

The dataset of pass or fail in an exam of 5 students is given in the following table. Use logistic regression as classifier and answer the following questions:

(i) Calculate the probability of pass for the student who studied 33 hours.

(ii) How many hours must a student study to have a greater than 95% probability of passing the course?

Assume the model suggested by the optimizer for odds of passing the course is, log (odds) = z = -64+2*hours.

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

Q.4(b) Consider the following dataset. Use ID3 algorithm to construct a decision tree for the classification of tuples. Clearly discuss each step of the classification process. [5] 4 2,5

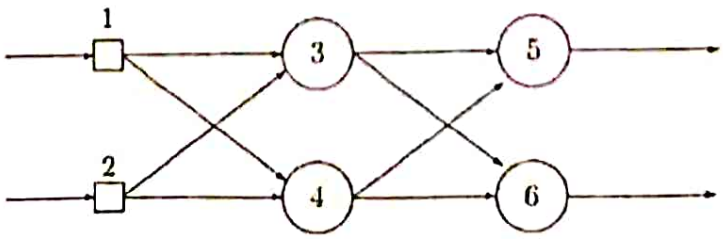
Altitude	Wind	Temperature	Outcome
High	Low	Hot	Crash
Low	High	Cold	Safe
Low	Low	Mild	Safe
Medium	High	Hot	Crash
High	Low	Mild	Safe
Medium	High	Mild	Crash
High	Low	Cold	Crash
Low	Low	Cold	Safe
Medium	Low	Mild	Safe
Low	High	Hot	Crash

Q.5(a) What is partitional clustering? Consider the table below and show the final step of agglomerative clustering with single linkage and complete linkage by a dendrogram. You need to show each step of the clustering process. [1+4=5] 5 2,5

	A	B	C	D	E	F
A	0					
B	0.12	0				
C	0.51	0.25	0			
D	0.84	0.16	0.14	0		
E	0.28	0.77	0.70	0.45	0	
F	0.34	0.61	0.93	0.20	0.67	0

Q.5(b) What is the significance of learning rate in artificial neural networks? [2+3=5] 5 2,5

The following diagram represents a feed-forward neural network with one hidden layer:



PTO

A weight on connection between nodes i and j is denoted by wij, such as w13 is the weight on the connection between nodes 1 and 3. The following table lists all the weights in the network:

$w_{13} = -2$ $w_{23} = 3$	$w_{35} = 1$ $w_{45} = -1$
$w_{14} = 4$ $w_{24} = -1$	$w_{36} = -1$ $w_{46} = 1$

Each of the nodes 3, 4, 5 and 6 uses the following activation function:

$$\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where v denotes the weighted sum of a node. Each of the input nodes (1 and 2) can only receive binary values (either 0 or 1). Calculate the output of the network (y5 and y6) for each of the input patterns:

Pattern:	P_1	P_2	P_3	P_4
Node 1:	0	1	0	1
Node 2:	0	0	1	1

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